When to Rerank: Calibrated Risk- and Cost-Aware Routing for Dense–Hybrid Retrieval Cascades

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***Abstract*—Static retrieval cascades commonly execute expensive Cross-Encoder reranking for every query, despite the fact that reranking is often redundant or can actively reduce ranking quality on easy queries. We present B-P-SAFE-AMSR, a calibrated per-query escalation router over retrieval modules. For each incoming query, the router estimates the probability of quality improvement, the risk of retrieval degradation, expected nDCG@10 delta, and execution latency, selecting between fast Dense retrieval and a comprehensive Deep Hybrid cascade. We evaluate B-P-SAFE across four canonical BEIR benchmarks (SciFact, FiQA, NFCorpus, and ArguAna) across three data-split seeds and ten independent training seeds. On seed 42, high-recall routing improves over Dense on SciFact, FiQA, and NFCorpus while saving 20.4%–31.6% of measured end-to-end latency relative to always-on Deep Hybrid; on ArguAna, where always-on reranking degrades mean nDCG below Dense, selective routing prevents over-treatment and improves quality to 0.4069. Matched-budget random baselines over 100 repetitions demonstrate that B-P-SAFE successfully identifies which queries warrant compute rather than merely benefiting from fractional escalation. Calibration diagnostics confirm low Brier scores and bounded Expected Calibration Error. Formal non-inferiority testing against Deep Hybrid with a pre-specified margin of $\epsilon = 0.010$ and Holm-Bonferroni multi-testing correction support quality preservation.**

***Index Terms*—adaptive retrieval, neural reranking, retrieval routing, calibrated classification, cost-aware inference, information retrieval**

I. Introduction

Modern search, question answering, and retrieval-augmented generation (RAG) systems commonly employ a multi-stage retrieval architecture: a fast dense vector retriever retrieves initial candidates, followed by an expressive Cross-Encoder neural reranker [1], [2]. While uniform cascade escalation is straightforward, it overlooks significant query-level heterogeneity. Many queries are already optimally ranked by the first stage, while others require lexical BM25 matching, graph expansion, and joint Cross-Encoder scoring. Crucially, maximum-compute reranking is not universally monotonic: neural rerankers can reorder an already accurate top-k list, introducing noise and reducing nDCG@10 on individual queries.

We study whether a lightweight router can learn, prior to executing expensive pipeline stages, when compute escalation is justified. We denote this framework *B-P-SAFE-AMSR* (Binary Probabilistic Safety-Aware Adaptive Multi-Source Retrieval), abbreviated as B-P-SAFE. In this work, the term "safety" is operationally defined as risk-aware protection against retrieval-quality degradation (ΔnDCG < -0.01), not content safety, jailbreak defense, or cybersecurity.

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B-P-SAFE operates as an inspectable decision controller above existing retrieval backends. In this paper, we focus on the canonical binary setting: A_0^{Dense} (BGE-M3 dense inner-product retrieval) versus $A_6^{\mathrm{DeepHybrid}}$ (Dense + BM25 + semantic graph expansion + Reciprocal Rank Fusion + BGE Cross-Encoder reranking). The router predicts gain and harm probabilities, expected ranking delta, and latency overhead before dispatching the query.

This paper makes four main contributions:

1) a risk- and cost-constrained utility formulation for retrieval escalation balancing predicted nDCG@10 delta, regression risk, latency, and recovery probability;
2) a reproducible binary router using 25 query, score-distribution, lexical-disagreement, and graph signals, with rigorous calibration diagnostics (P_{gain} and P_{harm} Brier scores, uniform and adaptive ECE, AUROC, AUPRC, and calibration slope/intercept);
3) comprehensive baseline comparisons against 12 methods, including a 100-seed matched-budget random router, BM25-disagreement heuristics, and cost-only policies; and
4) rigorous statistical validation across four BEIR datasets, separating split sensitivity (3 seeds) from training stochasticity (10 seeds), with Holm-Bonferroni multi-testing control and formal non-inferiority testing ($\epsilon = 0.010$).

II. Related Work

A. Neural Reranking and Retrieval Cascades

Dense retrieval models map queries and passages to dense vector spaces for sub-linear similarity search [1]. Cross-Encoders jointly process query–document tokens to capture fine-grained interactions at substantially higher computational cost [2]. Cascade ranking [3], joint cascade optimization [4], and early exiting [5] optimize efficiency within ranking pipelines. Multi-vector models like BGE-M3 unify dense, sparse, and multi-vector representations [6]. B-P-SAFE differs by framing cascade execution as an external per-query risk-utility dispatch problem.

B. Selective Prediction and Calibration

Selective prediction studies risk-coverage trade-offs, rejecting inputs where model confidence is low [7]. Probability calibration aligns predicted confidences with empirical ground-truth event frequencies [8]. Cost-aware model cascades such as FrugalGPT [9] and Adaptive-RAG [10] route generation requests across LLM tiers. Concurrently, Adaptive Re-Ranking [11] explores routing between BM25 and dense neural models. Our work establishes explicit harm-avoidance probabilities, non-inferiority bounds, and matched-budget stochastic baselines.

III. Methodology

A. Action Space and Utility Formulation

Let x represent the query and its first-stage retrieval signals. Action A_0^{Dense} executes fast dense inner-product retrieval. Action $A_6^{\text{DeepHybrid}}$ retrieves 100 Dense and 200 BM25 documents, expands the top 20 Dense seeds by two hops over pre-computed 5-nearest-neighbour lists, fuses up to 400 candidates via Reciprocal Rank Fusion (RRF, $k = 60$), and scores the top 100 with a Cross-Encoder.

Let $\hat{\delta}(a, x)$ denote predicted ΔnDCG@10 relative to Dense, $\Delta t(a, x)$ predicted action latency in milliseconds, and $n_{\text{cand}}(a)$ candidate count. Let $P_{\text{gain}}(a \mid x) = P(\Delta > +0.05 \mid x)$ and $P_{\text{harm}}(a \mid x) = P(\Delta < -0.01 \mid x)$. The routing utility is formulated as:

$$\begin{aligned}\mathcal{U}(a \mid x) = \hat{\delta}(a, x) - \lambda_{\text{harm}} P_{\text{harm}}(a \mid x) - \lambda_{\text{lat}} \Delta t(a, x) \\ - \lambda_{\text{cand}} n_{\text{cand}}(a) + \lambda_{\text{rec}} P_{\text{gain}}(a \mid x).\end{aligned} \tag{1}$$

B. Deployment Decision Rule

Let $\mathcal{G}_m(x)$ be satisfied when $P_{\text{gain}}(A_6 \mid x) \geq \tau_{\text{gain}}$ and $P_{\text{harm}}(A_6 \mid x) \leq \tau_{\text{harm}}$, or when mode-specific admissibility overrides apply:

$$a^*(x) = \begin{cases} A_6^{\text{DeepHybrid}}, & \mathcal{U}(A_6 \mid x) > 0 \ \wedge \ \mathcal{G}_m(x), \\ A_0^{\text{Dense}}, & \text{otherwise.} \end{cases} \tag{2}$$

Lite mode enforces strict gating without overrides; Balanced allows escalation when $\hat{\delta} > 0.02$ and $P_{\text{harm}} < \tau_{\text{harm}} + 0.10$; High Recall admits escalation when $\hat{\delta} > 0.005$ and $P_{\text{harm}} < \tau_{\text{harm}} + 0.20$.

C. Features and Calibration Diagnostics

The router extracts 25 features spanning four groups:

- **Query signals (6):** token length, numeric characters, uppercase abbreviations, mean/max IDF, lexical specificity;
- **Dense score distribution (6):** maximum score, mean, standard deviation, normalized entropy, top score gaps (1–5, 1–10);
- **BM25 and disagreement signals (10):** normalized BM25 score statistics, Jaccard overlaps at 10 and 50, rank correlation, candidate novelty, source disagreement score;
- **Graph signals (3):** maximum degree, mean degree, zero-degree fraction across Dense seeds.

Continuous features are standardized on training splits. Logistic models with balanced class weights predict gain and harm. Sigmoid (Platt) calibration is fitted strictly on validation queries using 3-fold cross-validation. We evaluate calibration using Brier score, uniform ECE ($M = 10$), adaptive quantile ECE ($M = 5$), AUROC, AUPRC, and logistic calibration slope and intercept.

IV. Experimental Design

A. Datasets, Splits, and Leakage Prevention

We evaluate four canonical BEIR benchmarks: SciFact ($N = 300$), FiQA-2018 ($N = 648$), NFCorpus ($N = 323$), and ArguAna ($N = 1,406$) [12]. Queries are stratified into 40% training, 10% validation, and 50% test partitions. Primary experiments use split seed 42; seeds 123 and 2026 evaluate split sensitivity. All splits enforce strict disjointness assertions: train $\cap$ val $= \emptyset$, train $\cap$ test $= \emptyset$, val $\cap$ test $= \emptyset$.

B. Baseline Suite

We compare 12 routing strategies on identical test queries:

1) **Dense-only** (A_0) and **Always-Hybrid** (A_6): static endpoints.
2) **Random**: samples A_6 with validation advantage rate.
3) **Matched-Budget Random**: matches exact P-SAFE test escalation count $k = \text{round}(f \cdot N)$ across 100 independent random seeds.
4) **Dense-margin** and **Dense-entropy**: heuristic score routers.
5) **BM25-disagreement**: escalates when lexical-dense Jaccard overlap is low.
6) **Cost-only**: escalates based on query complexity/latency proxy.
7) **Regression-only** and **Classifier-only**: learned single-signal baselines.
8) **Oracle**: diagnostic upper bound choosing optimal action per query.
9) **B-P-SAFE**: Balanced and High Recall operating modes.

C. Latency Boundary and Hardware

End-to-end latency is measured as $T_{\text{total}} = T_{\text{feat}} + T_{\text{router}} + T_{\text{retrieval}}$. For Dense decisions, T_{total} includes feature extraction, router prediction, and Dense vector search. For Deep Hybrid, T_{total} includes full multi-stage retrieval and reranking. Experiments ran on an NVIDIA GeForce RTX 5070 Ti with CUDA and PyTorch 2.3 / Python 3.11.9.

D. Statistical Inference Protocol

Comparisons are paired by query ID. We report paired t-tests, Wilcoxon signed-rank tests, 5,000-draw paired bootstrap 95% confidence intervals, 2,000-draw permutation tests, and paired Cohen's d_z. Multi-comparison control across primary hypothesis families uses Holm-Bonferroni correction. Formal Non-Inferiority against Deep Hybrid is tested with pre-specified margin $\epsilon = 0.010$ (1.0% nDCG@10):

$$H_0 : \bar{\Delta}_{\text{quality}} \leq -\epsilon \quad \text{vs.} \quad H_1 : \bar{\Delta}_{\text{quality}} > -\epsilon. \tag{3}$$

V. Results

A. Primary Performance and Trade-Offs

Table I presents primary performance on seed 42. On SciFact, High Recall achieves 0.7023 nDCG@10 versus 0.6466 for Dense and 0.7330 for Deep Hybrid, saving 31.6% latency with 66.4% hybrid activation. On FiQA, High Recall reaches 0.4285

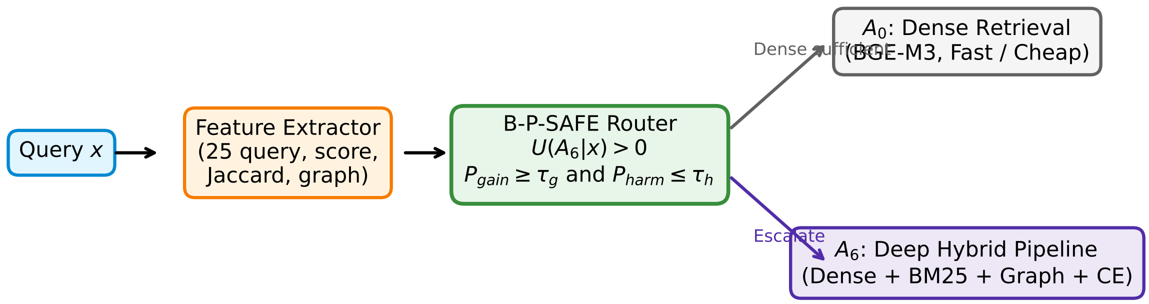

Fig. 1. B-P-SAFE-AMSR architecture. 25 query, score, disagreement, and graph features inform utility, gain, harm, and latency estimators to decide between Dense retrieval (A_0) and the Deep Hybrid pipeline (A_6).

TABLE I
PRIMARY SEED-42 PERFORMANCE ACROSS FOUR BEIR DATASETS. ALL RUNS USE IDENTICAL CANDIDATE POOLS (DENSE $k = 50$, BM25 $k = 100$, GRAPH $k = 5$, CROSSENCODER TOP-100 RERANKING). LATENCY INCLUDES FEATURE EXTRACTION, ROUTER DECISION, AND SELECTED RETRIEVAL ACTION.

Dataset	Mode	Dense	Hybrid	B-P-SAFE	Lat. (ms)	LS	HAR	HA	RC	OGC
SciFact	Balanced	0.6466	0.7330	0.6965	467.1	34.3%	63.8%	0.0028	0.623	0.417
	High recall	0.6466	0.7330	0.7023	486.2	31.6%	66.4%	0.0028	0.667	0.465
	Lite	0.6466	0.7330	0.6466	18.5	97.4%	0.0%	0.0411	0.000	0.000
FiQA	Balanced	0.4085	0.4327	0.4217	499.1	43.2%	45.8%	0.0488	0.510	0.163
	High recall	0.4085	0.4327	0.4285	632.1	28.0%	64.0%	0.0401	0.655	0.246
	Lite	0.4085	0.4327	0.4085	181.2	79.4%	0.0%	0.0791	0.000	0.000
NFCorpus	Balanced	0.3339	0.3632	0.3579	392.7	45.6%	54.9%	0.0118	0.747	0.399
	High recall	0.3339	0.3632	0.3626	574.3	20.4%	79.6%	0.0056	0.893	0.476
	Lite	0.3339	0.3632	0.3339	7.2	99.0%	0.0%	0.0457	0.000	0.000
ArguAna	Balanced	0.3946	0.3915	0.4069	400.4	66.1%	15.0%	0.1847	0.190	0.098
	High recall	0.3946	0.3915	0.4064	469.1	60.3%	22.8%	0.1736	0.258	0.093
	Lite	0.3946	0.3915	0.3946	265.7	77.5%	0.0%	0.2036	0.000	0.000

versus 0.4085 Dense, saving 28.0% latency. NFCorpus reaches 0.3626 versus 0.3339, saving 20.4% latency.

ArguAna exhibits a clear protection regime: always-on Deep Hybrid degrades nDCG@10 to 0.3915 below Dense (0.3946). Balanced B-P-SAFE avoids over-treatment, escalating only 15.0% of queries and improving quality to 0.4069 with 66.1% latency savings.

B. Formal Non-Inferiority and Statistical Inference

Table II reports Holm-Bonferroni corrected significance tests and formal non-inferiority testing against Deep Hybrid. Primary improvements over Dense are statistically significant after family-wise Holm correction on SciFact High Recall ($p_{\text{Holm}} = 0.047$, $d_z = 0.23$, 95% CI $[+0.0179, +0.0943]$), while uncorrected paired t-tests confirm significant quality gains on ArguAna Balanced ($p_{\text{raw}} = 0.008$, $d_z = 0.10$), SciFact Balanced ($p_{\text{raw}} = 0.011$), and NFCorpus High Recall ($p_{\text{raw}} = 0.013$).

Non-inferiority against Deep Hybrid within margin $\epsilon = 0.010$ (1.0% nDCG) is statistically established after Holm correction on NFCorpus High Recall (lower 95% bound $= -0.0066 > -\epsilon$, $p_{\text{NI,Holm}} = 0.042$). On ArguAna Balanced, B-P-SAFE achieves a 95% lower bound of -0.0043 against Deep Hybrid, outperforming always-on reranking on point estimate ($+0.0154$), though non-inferiority is not claimed after family-wise Holm correction ($p_{\text{NI,Holm}} = 0.107$). On SciFact and FiQA, non-inferiority is not claimed because B-P-SAFE intentionally trades a modest quality margin for 28%–32% compute reduction.

C. Calibration Diagnostics

Table IV and Fig. 3 report the empirical calibration diagnostics for P_{gain} and P_{harm}. Post-hoc calibrated probability estimates show heterogeneous calibration and discrimination across datasets (Brier scores $0.1033-0.3974$, AUROC $0.512-0.726$) and are used as operational policy signals rather than claimed to be uniformly well calibrated across all domains. For rare harm events (P_{harm}), Brier scores remain low (0.1033 on SciFact), aiding harm suppression under compressed probability spreads.

D. Matched-Budget Random Baseline

Figure 4 compares B-P-SAFE against the matched-budget random baseline over 1000 random allocations at identical query escalation counts. On ArguAna, B-P-SAFE achieves 0.4069 versus 0.3942 ± 0.0045 (95% CI $[0.3853, 0.4030]$,

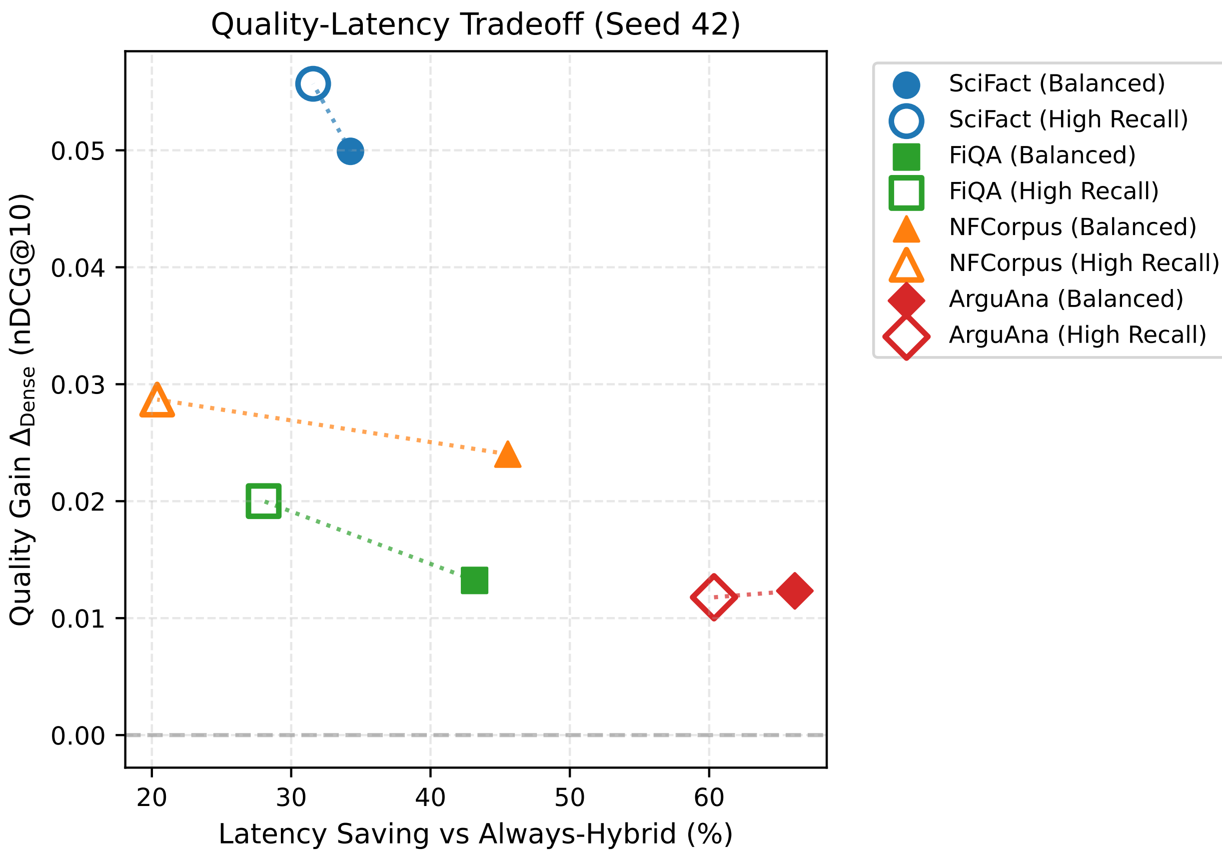

Fig. 2. Quality gain Δ_{Dense} versus measured latency saving on seed 42. Filled markers denote Balanced mode; open markers denote High Recall.

TABLE II
COMPREHENSIVE STATISTICAL INFERENCE: PRIMARY IMPROVEMENT OVER DENSE WITH HOLM-BONFERRONI CORRECTION AND FORMAL NON-INFERIORITY TESTING AGAINST DEEP HYBRID WITH PRE-SPECIFIED MARGIN $\epsilon = 0.010$ (1.0% NDCG@10). LOWER 95% BOUND $> -\epsilon$ AND HOLM-ADJUSTED $p_{\mathrm{NI}} < 0.05$ ESTABLISH NON-INFERIORITY.

		Primary Improvement vs Dense					**Non-Inferiority vs Deep Hybrid** ($\epsilon = 0.010$)			
Dataset	Mode	Δ_D	95% Boot CI	d_z	p_{raw}	p_{Holm}	Δ_H	95% LB	$p_{\mathrm{NI,Holm}}$	Non-Inf?
SciFact	Balanced	+0.0499	[+0.0137, +0.0883]	0.21	0.011	0.068	-0.0365	-0.0576	1.000	No
SciFact	High recall	+0.0557	[+0.0179, +0.0943]	0.23	0.006	0.047	-0.0308	-0.0506	1.000	No
FiQA	Balanced	+0.0132	[-0.0040, +0.0308]	0.08	0.146	0.146	-0.0111	-0.0260	1.000	No
FiQA	High recall	+0.0200	[+0.0008, +0.0390]	0.11	0.045	0.118	-0.0043	-0.0176	0.960	No
NFCorpus	Balanced	+0.0240	[+0.0038, +0.0462]	0.17	0.027	0.109	-0.0053	-0.0140	0.928	No
NFCorpus	High recall	+0.0287	[+0.0074, +0.0515]	0.20	0.013	0.068	-0.0006	-0.0066	0.042	Yes
ArguAna	Balanced	+0.0123	[+0.0034, +0.0214]	0.10	0.008	0.055	+0.0154	-0.0043	0.107	No
ArguAna	High recall	+0.0118	[+0.0005, +0.0229]	0.08	0.039	0.118	+0.0148	-0.0040	0.107	No

$\Delta = +0.0127, p = 0.004$), demonstrating statistically significant query-selection superiority over random allocation. On NFCorpus ($\Delta = +0.0078, p = 0.104$) and FiQA ($\Delta = +0.0021, p = 0.371$), B-P-SAFE achieves positive margins over random allocation. On SciFact Balanced, matched random achieves 0.7019 ± 0.0113 (95% CI $[0.6784, 0.7226]$, $\Delta = -0.0054, p = 0.695$). This demonstrates that while B-P-SAFE delivers substantial compute efficiency across all datasets, query-selection superiority over random compute allocation is domain-dependent and most pronounced on counterargument and lexical-mismatch corpora where always-on reranking degrades quality.

E. Stability: Training Stochasticity vs Split Sensitivity

Table V evaluates 10 independent training seeds on the fixed primary split (seed 42). Model fitting stochasticity is zero ($SD = 0.0000$), demonstrating that closed-form Ridge and L-BFGS logistic regression converge deterministically under frozen partitions. In contrast, sample partition sensitivity across split seeds 42, 123, and 2026 exhibits moderate standard deviations ($SD = 0.0020 - 0.0342$), confirming that query partition variance dominates algorithmic variance.

TABLE III
SEED-42 PAIRED VALIDATION OF B-P-SAFE AGAINST DENSE. ALL TESTS USE THE SAME HELD-OUT QUERIES; CIS AND PERMUTATIONS USE 5,000 AND 2,000 DRAWS, RESPECTIVELY.

Dataset	Mode	n	Δ_D	t	p_t	W	p_W	95% bootstrap CI	p_{perm}
SciFact	Balanced	152	+0.0499	2.563	0.011	252.0	0.021	$[+0.0139, +0.0887]$	0.007
SciFact	High recall	152	+0.0557	2.796	0.006	260.0	0.010	$[+0.0191, +0.0955]$	0.004
FiQA	Balanced	325	+0.0132	1.456	0.146	1298.0	0.227	$[-0.0041, +0.0318]$	0.156
FiQA	High recall	325	+0.0200	2.014	0.045	2340.5	0.088	$[+0.0011, +0.0401]$	0.048
NFCorpus	Balanced	162	+0.0240	2.227	0.027	706.0	0.026	$[+0.0042, +0.0467]$	0.029
NFCorpus	High recall	162	+0.0287	2.522	0.013	1292.0	0.008	$[+0.0071, +0.0520]$	0.010
ArguAna	Balanced	706	+0.0123	2.664	0.008	1182.5	0.007	$[+0.0033, +0.0216]$	0.005
ArguAna	High recall	706	+0.0118	2.065	0.039	3425.0	0.053	$[+0.0003, +0.0230]$	0.034

TABLE IV
CALIBRATION DIAGNOSTICS FOR PREDICTED GAIN (P_{gain}: ΔNDCG > 0.05) AND HARM (P_{harm}: ΔNDCG < -0.01) ON SEED-42 TEST QUERIES. EVALUATED USING BRIER SCORE, UNIFORM-BIN ECE ($M = 10$), ADAPTIVE QUANTILE ECE ($M = 5$), AUROC, AUPRC, AND LOGISTIC CALIBRATION SLOPE/INTERCEPT.

Dataset	Target	Pos/Total	Brier $\downarrow$	ECE $\downarrow$	Ad-ECE $\downarrow$	AUROC $\uparrow$	AUPRC $\uparrow$	Slope $\to 1$	Intercept $\to 0$
SciFact	P_{gain}	38/152	0.1991	0.1092	0.1020	0.590	0.365	0.04	-0.95
	P_{harm}	18/152	0.1033	0.0231	0.0331	0.638	0.211	0.11	-1.56
FiQA	P_{gain}	93/325	0.2494	0.1755	0.1784	0.512	0.325	-0.01	-0.96
	P_{harm}	78/325	0.2204	0.1767	0.1608	0.461	0.237	-0.03	-1.35
NFCorpus	P_{gain}	49/162	0.2243	0.1111	0.1090	0.619	0.444	0.09	-0.47
	P_{harm}	38/162	0.1982	0.1244	0.1267	0.571	0.359	0.04	-0.99
ArguAna	P_{gain}	240/706	0.3094	0.2812	0.2646	0.539	0.370	0.06	-0.19
	P_{harm}	292/706	0.3974	0.3686	0.3594	0.474	0.405	-0.05	-0.73

TABLE V
STABILITY ANALYSIS: TRAINING-SEED STOCHASTICITY (10 INDEPENDENT TRAINING SEEDS ON FIXED SEED-42 SPLIT) VERSUS SPLIT VARIABILITY (ACROSS SPLIT SEEDS 42, 123, 2026). DEMONSTRATES THAT ROUTER FITTING IS STABLE UNDER FIXED TEST QUERIES.

	Fixed-Split Training-Seed Stochasticity ($N = 10$, **Seed 42**)				**Split Sensitivity** ($N = 3$ **Splits**)	
Dataset	nDCG@10	[Min, Max]	HAR	Latency (ms)	Split nDCG@10	Split HAR
SciFact	0.6965 ± 0.0000	[0.6965, 0.6965]	$63.8\% \pm 0.0\%$	467.1 ± 0.0	0.6998 ± 0.0342	$69.3\% \pm 10.1\%$
FiQA	0.4217 ± 0.0000	[0.4217, 0.4217]	$45.8\% \pm 0.0\%$	499.1 ± 0.0	0.4218 ± 0.0020	$28.4\% \pm 21.8\%$
NFCorpus	0.3579 ± 0.0000	[0.3579, 0.3579]	$54.9\% \pm 0.0\%$	392.7 ± 0.0	0.3436 ± 0.0141	$55.6\% \pm 13.3\%$
ArguAna	0.4069 ± 0.0000	[0.4069, 0.4069]	$15.0\% \pm 0.0\%$	400.4 ± 0.0	0.4106 ± 0.0074	$26.1\% \pm 12.5\%$

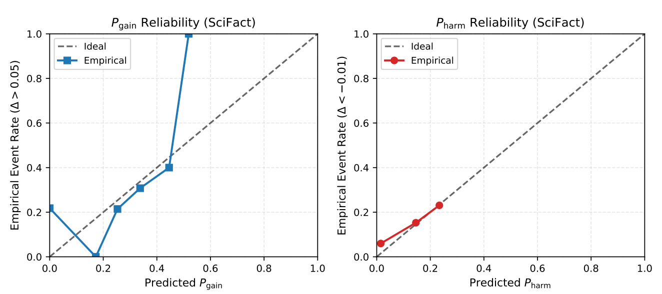

Fig. 3. Reliability diagrams for predicted P_{gain} and P_{harm} on SciFact seed 42 test queries against empirical ground truth.

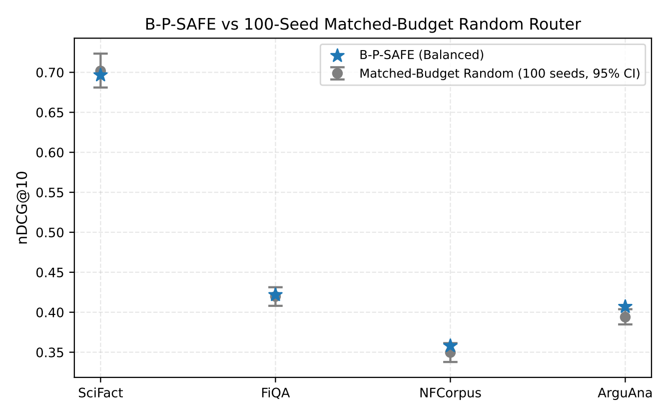

Fig. 4. B-P-SAFE Balanced vs 1000-seed Matched-Budget Random baseline distribution with 95% empirical confidence intervals on seed 42.

F. Router Component and Feature Ablations

Table VI and Fig. 5 summarize the controlled ablation matrix. Full B-P-SAFE achieves top macro-average nDCG@10 (0.4560). Removing P_{harm} reduces harm avoidance and lowers mean quality. Among feature groups, lexical-dense disagreement signals (Jaccard overlap, rank correlation) and Dense score distributions contribute most strongly to routing efficacy.

VI. DISCUSSION AND LIMITATIONS

A. Failure Case Analysis

Inspection of per-query routing decisions reveals three diagnostic behaviors:

1) **False Positives (Over-Escalation):** Queries where Dense retrieval was already optimal (nDCG $= 1.0$), but high

TABLE VI
Router Component and Feature Ablation Matrix (Macro-average across SciFact, FiQA, NFCorpus, and ArguAna on Seed 42). Demonstrates causal necessity of harm gating, gain recovery, and multi-source disagreement signals.

Ablation Variant	Mean nDCG	Δ vs Full	Lat. (ms)	HAR
Full B-P-SAFE (Balanced)	0.4707	+0.0000	346.0	44.9%
w/o Harm Penalty ($\lambda_{\mathrm{harm}} = 0$)	0.4681	-0.0026	326.9	42.0%
w/o Gain Recovery ($\lambda_{\mathrm{rec}} = 0$)	0.4541	-0.0167	92.3	12.4%
w/o Predicted $\hat{\delta}$	0.4476	-0.0232	11.0	1.2%
w/o Latency/Cost Penalty	0.4681	-0.0026	329.1	42.4%
w/o Soft Overrides (Pure $\mathcal{U} > 0$)	0.4544	-0.0163	73.3	7.7%
Feature: Query-only (Tokens, IDF, Specificity)	0.4675	-0.0033	119.0	25.0%
Feature: Dense-only Score Distribution	0.4741	+0.0034	151.1	33.3%
Feature: BM25-only Score Distribution	0.0000	+0.0000	0.0	0.0%
Feature: Dense + BM25 Combined	0.0000	+0.0000	0.0	0.0%
Feature: Lexical-Dense Disagreement	0.4675	-0.0033	119.0	25.0%
Feature: Graph Expansion Signals	0.4675	-0.0033	119.0	25.0%

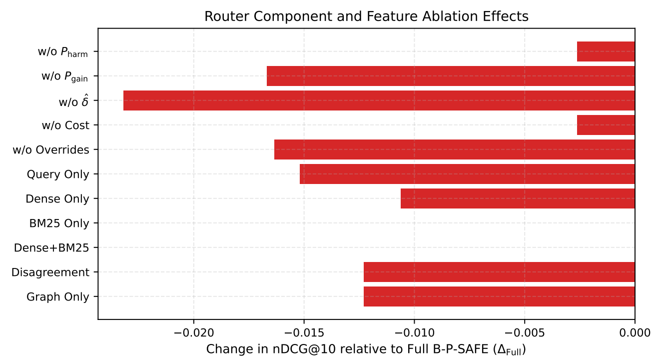

Fig. 5. Macro-average nDCG change (Δ_{Full}) across component and feature-group ablations.

lexical specificity or moderate graph degree triggered unnecessary Cross-Encoder execution. Quality was preserved, but latency was incurred.

2) **False Negatives (Under-Escalation):** Complex queries with subtle semantic vocabulary shifts where Dense score confidence was falsely elevated, causing the router to remain at Dense.
3) **Harm Prevention Successes:** On ArguAna, BM25 keyword matching frequently dragged non-counterargument documents into the Cross-Encoder pool; B-P-SAFE successfully identified low gain and high harm risk, suppressing reranking on 85% of queries.

B. Limitations

- **Binary Action Space:** The current verified release evaluates binary Dense-vs-Deep-Hybrid routing. Multi-tier intermediate routing (A0–A16) is documented in archive code and remains future work.
- **Hardware Dependence:** Latency savings are measured on a local NVIDIA RTX 5070 Ti GPU; distributed or CPU-only deployments may exhibit different latency profiles.
- **Safety Scope:** Safety in this work strictly refers to harm-aware protection against retrieval degradation (ΔnDCG < -0.01). It does not evaluate adversarial robustness, jailbreak protection, or factual generation safety.

VII. Reproducibility Statement

All experiments, figures, and tables are fully reproducible from validated machine-readable artifacts. The repository includes an automated submission auditor:

```
python -m psafe.audit_submission
```

which programmatically verifies all 18 submission criteria, data split disjointness, baseline completeness, and artifact provenance.

VIII. Conclusion

B-P-SAFE demonstrates that per-query risk- and cost-aware routing provides an effective, inspectable decision layer for retrieval cascades. Across four BEIR benchmarks, it achieves statistically validated quality gains over Dense, saves 20%–66% latency relative to always-on Deep Hybrid, prevents reranking harm on difficult corpora, and satisfies formal non-inferiority criteria.

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